

$$M(4.0 \cos 50^\circ) = 45 (7.0 + 4.0 \cos 50^\circ) \Rightarrow M = 170 \text{ N}$$

- 8 D** GPE gained =  $mgh = mg \sin \alpha .s$   
 Work done by force =  $F s$   
 Divide the 2 eqns give answer D.

**9 D**  $F \sin \theta = 10 \sin 30^\circ = 5 \text{ N}$   $F \cos \theta = 10 \cos 30^\circ = 8.66 \text{ N}$

$2 \text{ N}$   $F \sin \theta = 10 \sin 30^\circ = 5 \text{ N}$